

Ideal Sampling

- $x(t)$ is a low pass signal (a smooth function)
- W is its bandwidth (all frequencies describing $x(t)$)
- Ideal sampling is performed by multiplying $x(t)$ with a Dirac delta train

$$x_s(t) = x(t) \times \sum_{k=-\infty}^{+\infty} \delta(t - kT_s)$$

where T_s is a sampling interval.

Illustration

- Signal as a function of time:

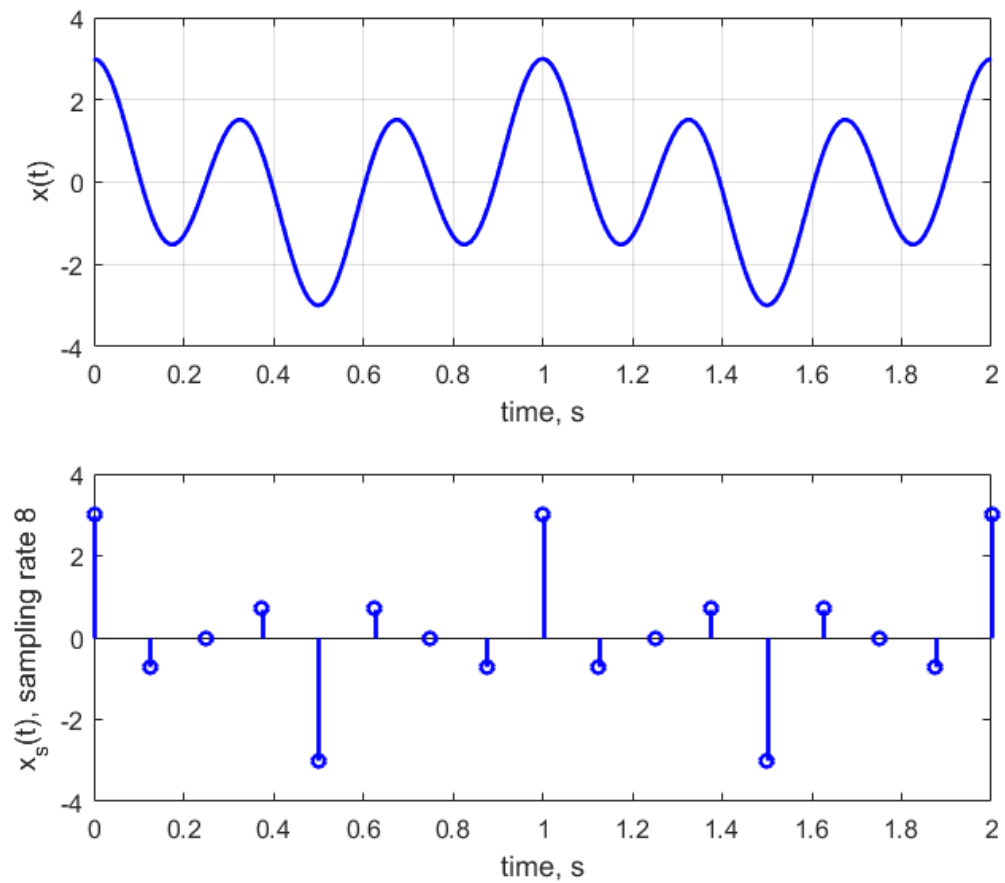
$$x(t) = \cos(2\pi t) + 2\cos(6\pi t)$$

- Its Fourier transform:

$$X(f) = \frac{1}{2}\delta(f \pm 1) + \delta(f \pm 3)$$

- If we treat it as a low pass signal (described by low frequencies), then its bandwidth is $W = 3$.
- Let us sample the signal at the rate $f_s = 8$ Hz (the sampling period is $T_s = 1/8$ s).

Illustration



FT of sampled signal

- FT of the sampled signal:

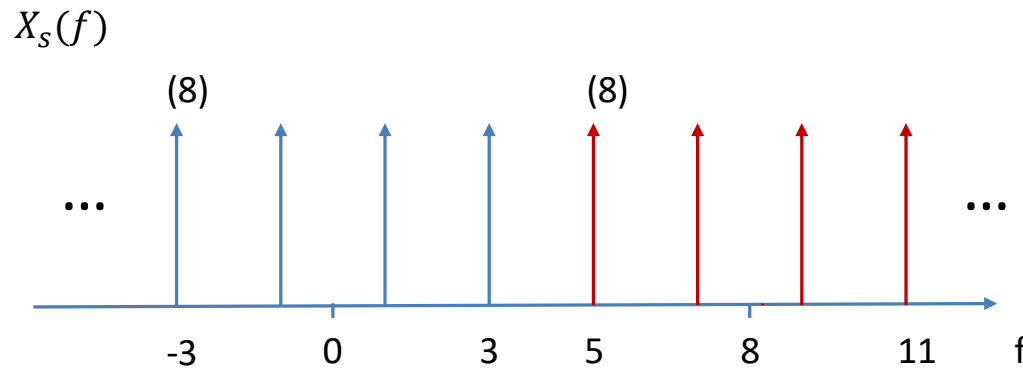
$$X_s(f) = X(f) * \sum_{k=-\infty}^{+\infty} f_s \delta(f - kf_s)$$

“*” stands for correlation and $f_s = 1/T_s$

- It can also be written as:

$$X_s(f) = \sum_{k=-\infty}^{+\infty} f_s X(f - kf_s)$$

Many copies of $X(f)$!



FT of the sampled signal

$$x(t) = \cos(2\pi t) + 2\cos(6\pi t)$$

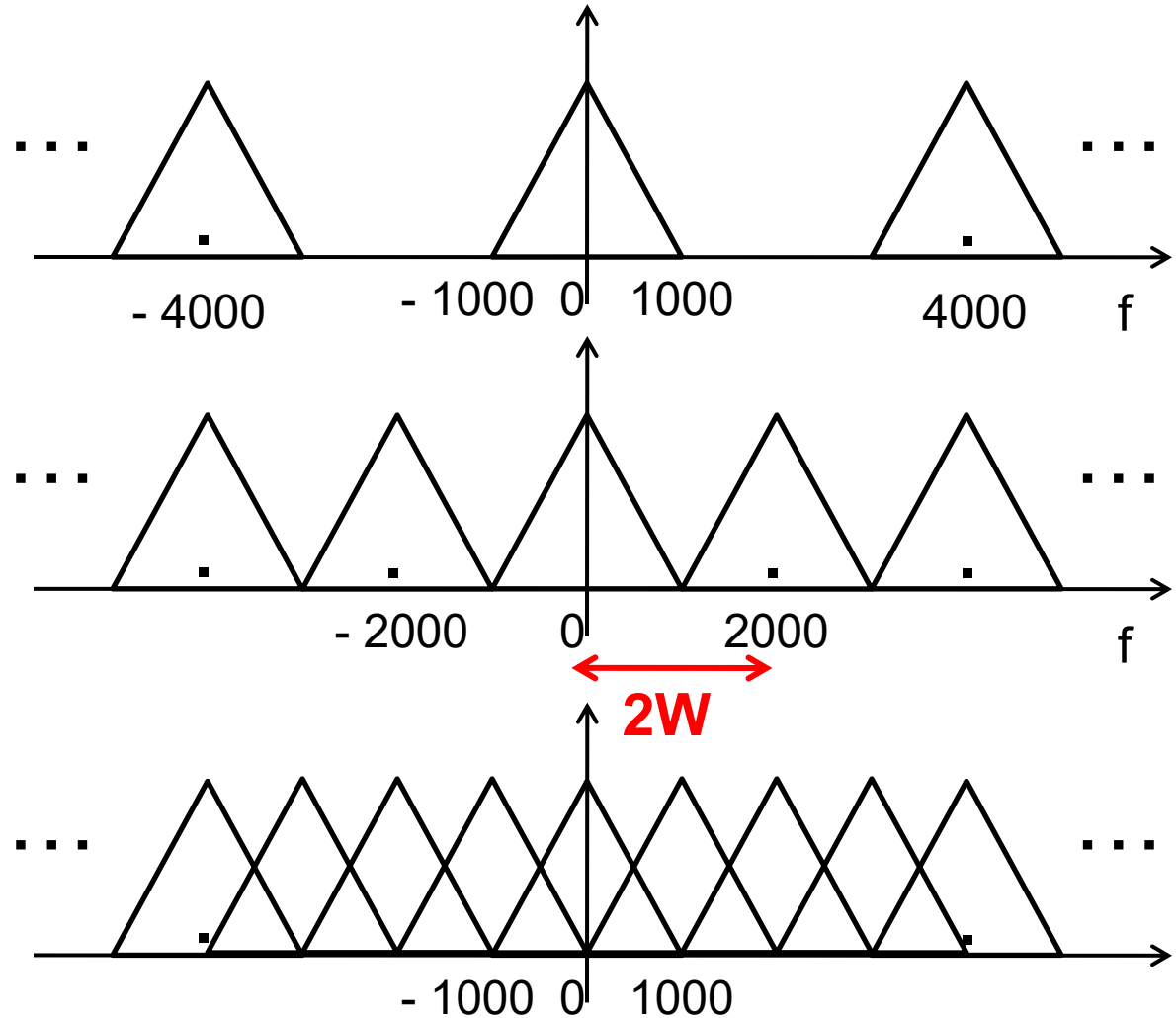
sampled at the rate $f_s = 8$ Hz is

$$X_s(f) = \sum_{k=-\infty}^{+\infty} (4\delta(f \pm 1 - k8) + 8\delta(f \pm 3 - k8))$$

Back to
illustration

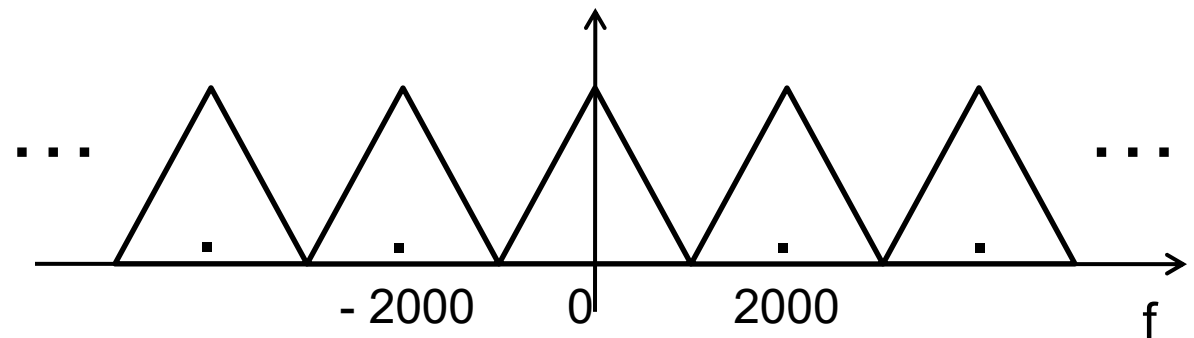
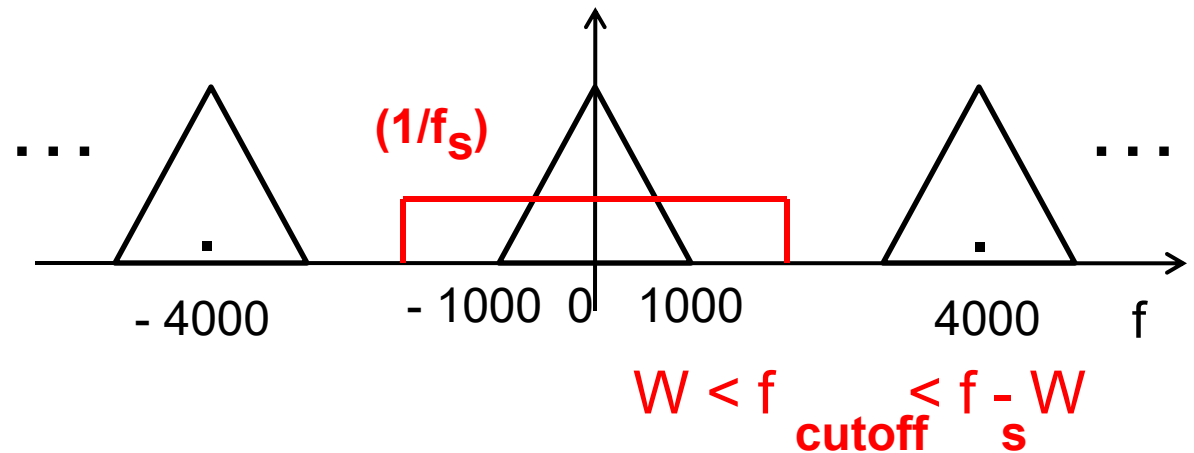
Nyquist Sampling Theorem

- $W=1000$ is the bandwidth of a band pass signal
- Find the smallest value of sampling rate such that spectral copies do not overlap.

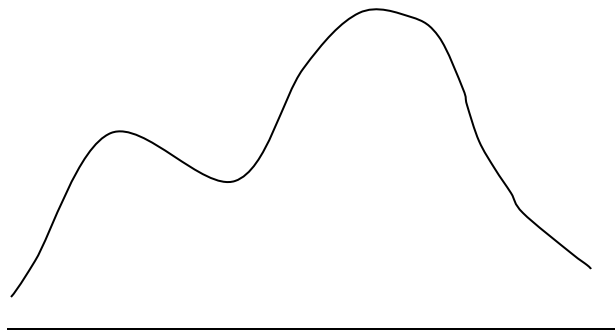


Nyquist Sampling Theorem

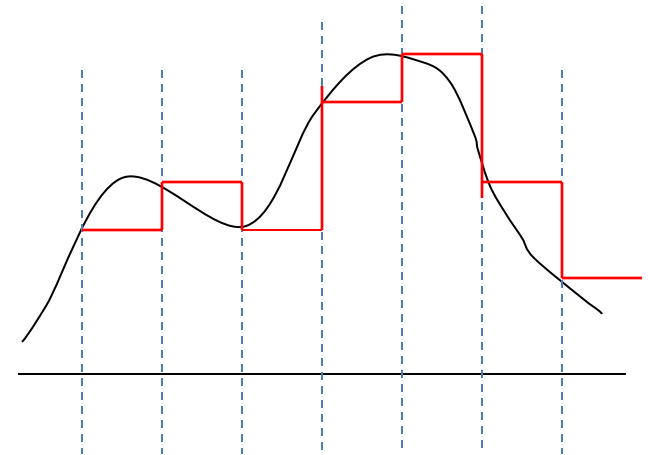
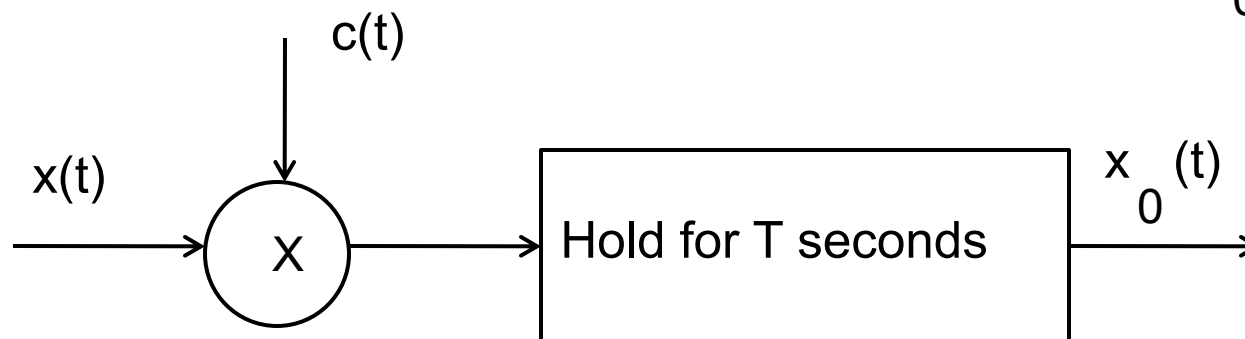
- Assume that sampling frequency is high (spectral copies do not overlap)
- How to recover the original spectrum?



Sampling with Zero-Order Hold

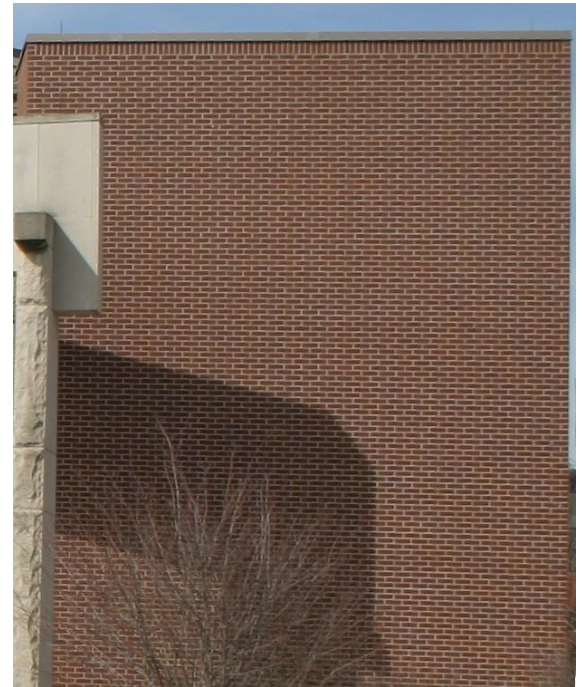
 $x(t)$

Zero-order
hold

 $x_0(t)$ 

Aliasing

- Left panel: Properly Sampled image of brick wall (493-by-599)
- Right panel: Spatial aliasing in the form of a [Moiré pattern](https://en.wikipedia.org/wiki/Moir%C3%A9_pattern) (205-by-250)
- <http://en.wikipedia.org/wiki/Aliasing>



Aliasing (another example)



Aliasing due to undersampling

Application of anti-aliasing filter

